SWAJAL-WICK

Zero-Energy Capillary Root Irrigation System

Target Application: Drought-Prone Agricultural Regions (e.g., North Karnataka, Kolar, Tumakuru)
Core Principle: Capillary Action / Sub-Surface Irrigation
Estimated Cost: < ₹50 per plant installation (utilizing recycled materials)

1. Introduction & Problem Statement

In water-scarce regions of Karnataka, smallholder farmers struggle to maintain crop yields due to unpredictable monsoons, depleting groundwater, and high evaporation rates. Traditional surface irrigation methods result in massive water losses (up to 50%) due to runoff, deep percolation, and direct solar evaporation. While conventional drip irrigation significantly mitigates these losses, it presents insurmountable barriers for marginal farmers: high initial capital costs, dependence on continuous grid electricity or expensive solar pumps for pressurization, and severe issues with emitter clogging from hard water salts and silt.

Swajal-Wick was developed to democratize efficient water management. It strips away the need for electricity, high-pressure pumps, and delicate filtration systems by utilizing natural physics.

2. Core Scientific Principle

The system operates purely on **capillary action**, which is the ability of a liquid to flow in narrow spaces without the assistance of, or even in opposition to, external forces like gravity. This occurs due to intermolecular forces.

> **The Physics of Capillary Flow:**
> Water movement through the wick is governed by the interplay of cohesion (water molecules sticking to each other) and adhesion (water molecules sticking to the wick fibers). The height h to which the capillary action will take the water is defined by Jurin's Law:
>
> $$h = (2\gamma \cos\theta) / (\rho g r)$$
>
> Where γ is the surface tension, θ is the contact angle, ρ is the density of water, g is acceleration due to gravity, and r is the radius of the capillary pores in the wick.

In the Swajal-Wick system, the plant acts as the regulator. As the plant transpires and the surrounding root-zone soil dries out, a moisture gradient is created. This gradient triggers the capillary wick to pull water from the buried reservoir precisely as needed. When the soil reaches optimal moisture, the capillary pull equalizes, and water transfer halts automatically. **It is a self-regulating, plant-driven system.**

3. System Architecture & Setup

The architecture is exceptionally simple, ensuring easy deployment by un-trained rural populations:

- **The Reservoir:** A 2-liter to 5-liter recycled PET bottle or small PVC pipe segment buried vertically next to the crop. Being sub-surface, it is insulated from solar heat, completely eliminating surface evaporation.
- **The Capillary Wick:** A braided rope (ideally a blend of highly absorbent cotton and rot-resistant synthetic fibers like nylon) inserted into the reservoir, with the other end buried directly into the active root zone of the plant.
- **The Interface:** The reservoir is refilled manually once a week through a small protruding neck above the soil surface, covered by a simple cap to prevent mosquito breeding and debris entry.

4. Material Sourcing & Cost Breakdown

A major focus of the Swajal-Wick design is utilizing locally available or upcycled materials to bring the cost barrier effectively to zero for rural farmers.

Component	Proposed Material Source	Estimated Cost (per unit)
Water Reservoir	Discarded PET bottles, bulk purchased from local scrap dealers.	₹2.00 - ₹5.00
Capillary Wick	Recycled fabric strips or bulk nylon-cotton braided ropes.	₹10.00 / meter
Refill Cap	Standard bottle caps or local clay stoppers.	₹0.00 (included)
Total Cost	**Ready for farm deployment**	**₹12.00 - ₹15.00**

5. Competitive Advantage & Impact

- **Zero Energy Requirement:** Eliminates the need for electric or diesel pumps. Operates passively 24/7.
- **Zero Clogging:** Unlike microscopic drip emitters, wicks do not clog with hard water calcification or silt, eliminating maintenance downtime.
- **Water Optimization:** Achieves nearly 100% water usage efficiency since water is delivered directly to the root membrane with zero surface evaporation.
- **Weed Suppression:** Because the surface soil remains completely dry, weed germination around the crop is drastically reduced, saving farmers hours of manual labor.